

Stock Market Wizard: Analyzing the Philippine Stock Market through Machine Learning Models

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Abstract—The stock market serves as a significant platform for investors, as it often reflects the financial health of the country. In the Philippines, the Philippine Stock Exchange (PSE) is central to stock trading and investment decisions. This study applied Linear Regression, Support Vector Machine (SVM), and Long Short-Term Memory (LSTM) models to historical data from both the Philippine Stock Exchange (PSE) and the Philippine Stock Exchange Index (PSEi). Results indicated that Linear Regression outperformed the other models in predictive accuracy. Additionally, the researchers generated a three-week forecast for stock prices using both datasets. Linear Regression and SVM successfully predicted price direction for up to four days on the PSE dataset and six days on the PSEi dataset before predictions diverged from actual values. The limitations highlight challenges regarding the external factors that can affect the stock prices like natural disasters. The models rely solely on historical patterns from the dataset and do not account for unforeseen events. While the models can generate accurate short-term predictions, it remains crucial to stay informed about external factors that may significantly affect stock prices.

Keywords—Stock Market, Machine Learning, Linear Regression, LSTM, SVM

I. INTRODUCTION

The stock market, also referred to as the equity market or share bazaar, is a marketplace where shares of publicly held companies are bought and sold [1]. It consists of exchanges where companies issue shares and other securities for trading. The stock market often uses the performance of several companies as a representation of the entire market since it is difficult to track the share value of each company individually [2].

It is important to keep track of the stock market's value, as it has become a key platform for businesses to raise capital and serves as an indicator of a country's financial health. Investors participate by buying stocks to gain profits as their value increases or by purchasing bonds for interest income [3].

In the Philippines, the Philippine Stock Exchange (PSE) is the central hub for stock trading. Understanding how the Philippine stock market operates can help you make informed decisions about investments, economic trends, and the country's overall financial environment [3].

One way to understand to predict stock market prices is through using machine learning algorithms. Stock market trend prediction is regarded as a crucial undertaking that requires careful consideration because, by making the right choices, accurate stock price predictions can result in substantial gains. Due to non-stationary, blatant, and chaotic data, stock market prediction is extremely difficult, making it difficult for investors to use their funds in a way that will maximize earnings [4].

II. REVIEW OF RELATED LITERATURE

With increasing competition in the market, it is important for investors to explore various forecasting methods. In a 2019 study by Ryll and Seidens, they compared and evaluated machine learning algorithms for forecasting against traditional stochastic methods. Their study concluded that machine learning algorithms outperformed these traditional methods [5]. They also concluded that SVM and ANN have the highest rank score results for the different algorithm classes.

Nabipour et al. (2020) predicted stock market trends using various machine learning and deep learning algorithms with both continuous and binary data. They employed the following machine learning algorithms: Decision Tree, Random Forest, Adaptive Boosting (AdaBoost), Extreme Gradient Boosting (XGBoost), Support Vector Classifier (SVC), Naïve Bayes, K-Nearest Neighbors (KNN), Logistic Regression, and Artificial Neural Network (ANN); along with two deep learning algorithms: Recurrent Neural Network (RNN) and Long Short-Term Memory (LSTM) [6].

In their study, they evaluated the performance of each algorithm using the F1-Score, Accuracy, and Receiver

Operating Characteristic Area Under the Curve (ROC-AUC). The algorithms were applied to four stock market sectors—petroleum, diversified financials, basic metals, and non-metallic minerals—over a ten-year period from November 2009 to November 2019. Their results showed that each algorithm performed well when continuous data was used, though performance improved when binary data was applied, with accuracy increasing from 67% to 83%. It was also revealed that RNN and LSTM outperformed the machine learning models [6].

In addition, Kakarla et al. (2024) predicted stock prices using the Long Short-Term Memory (LSTM) algorithm, Convolutional Neural Network (CNN), and Artificial Neural Network (ANN). They applied these three algorithms to the Indian stock market, specifically MAANG and Divis Labs, and found that the neural network-based ANN yielded the best results, as it captures intricate, non-linear relationships and patterns. Their study revealed that while all three algorithms performed well on the dataset, the LSTM outperformed the others, with Root Mean Square Error (RMSE) and accuracy scores of 0.42 and 0.55, respectively [7].

Furthermore, a study from Indika et al. (2021) compared several heterogeneous ensembles for financial forecasting, including averaging, weighted, stacking, and blending ensembles, and used a random forest regressor as the baseline. Regression was used to predict the next day's closing stock prices. The study used LSTM models, Linear Regression, and SVM as individual models; Analyzed 10 years of historical data of the 20 most active companies of the NASDAQ stock exchange for implementing ensemble models and found that SVM models underperformed, LSTM outputs were satisfactory, and Linear Regression models produced promising results as higher positive correlation exists between the target values and the training features, which has resulted in better performance of the linear regression model.

III. METHODOLOGY

A. Dataset

The dataset used for this study is from the Philippine Stock Exchange Index and Philippine Stock Exchange, covering a 10-year period ranging from October 13, 2014, to October 11, 2024. It includes four features: open, high, low, and volume and will be plotted based on their dates. The closing price of the stock will be predicted based on the features provided. Below is a description of each feature:

Open: The starting price of each share at the opening of the trade.

High: The highest price at which the stock traded during the trading day. This feature reflects the peak value of the stock within the day, giving insight into the stock's volatility and maximum potential value reached.

Low: The lowest price at which the stock traded during the trading day. It represents the minimum price point and can help in understanding the stock's risk and downside during the day.

Volume: The total number of shares traded during the day. High volume often indicates strong interest in the stock, which can correlate with significant price changes or investor activity.

Change: The difference between the current price and the previous trading day's closing price. It shows how much the price of the stock has increased or decreased as a percentage.

B. Algorithm

Linear Regression

Linear regression models fit a straight line or surface that minimizes the discrepancies between predicted and actual output values. There are simple linear regression calculators that use a "least squares" method to discover the best-fit line for a set of paired data. You then estimate the value of X (dependent variable) from Y (independent variable) [11].

Support Vector Machine (SVM)

The support vector machine (SVM) finds the best line or hyperplane in an N-dimensional space to maximize the distance between each class to classify data. Because it can manage both linear and nonlinear classification tasks, it is commonly employed in machine learning [10].

According to Chhajer et al. (2024) Support Vector Machines, or SVMs, are superior machine learning models that can be applied to problems involving regression or classification. The SVM approach, which is most applied to classification problems, performs optimally in high-dimensional spaces or in scenarios where the number of samples is less than or equal to multiple dimensions. This algorithm acts as a risk management tool and offers investors abnormal profits on their investments [9].

Long Short-Term Memory (LSTM)

Hochreiter & Schmidhuber's Long Short-Term Memory (LSTM) is an enhanced recurrent neural network design. They are excellent for jobs like speech recognition, language translation, and time series forecasting because they can identify long-term dependencies in sequential data [8].

Chhajer et al. (2024) explained that LSTM is capable of learning order dependence to solve sequence prediction issues. The forget gate, input, output, and cell make up the LSTM algorithm. The data entering and leaving the cell is managed by the gates. The fundamental building block of the model is the cell, which retains values for varied amounts of time [9].

IV. RESULTS AND DISCUSSION

A. Encountered Challenges

The researchers encountered several challenges in creating models for stock market predictions. One notable issue was the slower run time of the LSTM model in Google Colab compared to the other two algorithms used. Additionally, they faced problems with model overfitting.

In addition, stock market is an overly complex system, and it can overwhelm both human and computer's analysis towards it. Events like the pandemic and natural disasters, such as typhoons, can lead to discrepancies between the model's predictions and actual values. The datasets used for this study only predict the stock price based on historical data and patterns, and it may struggle to account for the unprecedented events.

B. Machine Learning Algorithm Evaluation

Philippine Stock Exchange			
	LR	SVM	LSTM
Mean Squared Error (MSE)	7.76	8.36	14.41
Mean Absolute Error (MAE)	2.05	2.22	2.98
Root Mean Squared Error (RMSE)	2.79	2.89	3.80
R-squared	0.9609	0.9575	0.9267
Philippine Stock Exchange Index			
	LR	SVM	LSTM
Mean Squared Error (MSE)	3956.55	3945.85	10981.80
Mean Absolute Error (MAE)	49.42	49.22	82.22
Root Mean Squared Error (RMSE)	62.90	62.82	104.79
R-squared	0.9526	0.9514	0.8647

Fig. 1. Evaluation Metrics

Linear Regression performed the best out of the three algorithms in both datasets used with an R-squared of 0.96 and 0.95 for PSE and PSEi, respectively. In addition, SVM outperformed LSTM in both datasets. This means that Linear Regression was able to predict the best, followed by SVM, and LSTM. Lastly, Linear Regression and LSTM performed

better on the PSE dataset while SVM performed better for the PSEi dataset.

C. Actual Value vs. Predicted Value

In addition to generating the evaluation metrics for the models used, the researchers also produced three-week predictions for the stock market, excluding weekends and holidays.

Philippine Stock Exchange

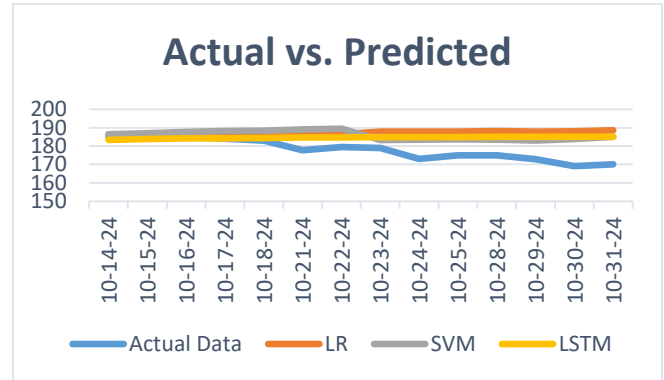


Fig. 2. Actual vs Predicted (PSE)

All three models show the same movement for the first six (6) day prediction, however a divergence emerged between the predicted and actual data after four (4) days. This discrepancy can be attributed to the model's reliance solely on historical data, without accounting for external factors like natural disasters. During the observation period, a typhoon impacted the Philippines, leading to a significant decline in Philippine stock market. This event underscores the model's limitation to adjusting in real life disruptions, which is beyond the scope of our historical pattern.

Philippine Stock Exchange Index

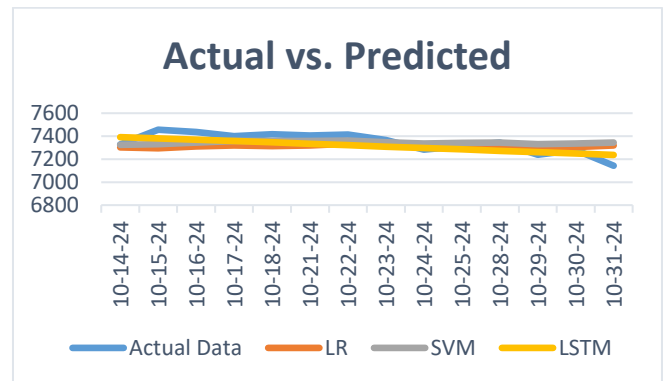


Fig. 3. Actual vs Predicted (PSEi)

Apart from a sudden peak in stock prices observed on the second day of prediction, both the Linear Regression (LR) and Support Vector Machine (SVM) models performed well in predicting the general direction of the Philippine Stock Exchange index (PSEi). However, the predictions diverged

from actual prices after the seventh day, as the actual prices showed a significant decline compared to the days before it. This discrepancy is similar to the one observed with the previous dataset since the discrepancy can be attributed to the model's dependency on historical data alone. As mentioned, during the observation period, the Philippines was hit by a typhoon, leading to the closure of businesses during the week causing a significant drop in the stock market prices.

V. CONCLUSION

This study used Linear Regression, Support Vector Machine (SVM), and Long Short-Term Memory (LSTM) to learn and predict the Philippine Stock Exchange (PSE) and Philippine Stock Exchange Index (PSEi). The results shows that Linear Regression was able to outperform SVM and LSTM on test data on both datasets used. It was followed by SVM and LSTM. This indicates that Linear Regression most effectively captured the patterns in historical data.

For the PSE dataset, the models' prediction was able to generate the same direction as the actual data up to four days, while up to six days for the PSEi dataset. However, it falls short as all models showed limitations in adjusting for unforeseen events, such as natural disasters, Linear Regression proved the most dependable for short-term trend prediction.

In line with this, the researchers emphasize the importance of caution and a thorough understanding that machine learning models, in this case, learn only from historical data and still have limitations. Investors should remain informed of other external factors, such as pandemics and natural disasters, which may also influence stock prices.

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